



Figure 4: A three-story building modeled as a mass-spring system.

$$\begin{aligned}
 \text{Mass 1: } m_1 x_1'' &= -K_1 x_1 + K_2 x_2 - K_2 x_1 \\
 \text{Mass 2: } m_2 x_2'' &= +K_3 x_3 - K_3 x_2 - k_2 x_2 + k_2 x_1 \\
 \text{Mass 3: } m_3 x_3'' &= -k_3 x_3 + K_3 x_2
 \end{aligned}$$

$$\begin{bmatrix}
 x_1 & x_2 & x_3 \\
 -k_1 - k_2 & k_2 & 0 \\
 k_2 & -k_2 - k_3 + k_3 & 0 \\
 0 & k_3 & -k_3
 \end{bmatrix}$$

$$\lambda = \omega^2$$